

A six-sevenths bound for transversals of hypergraphs with property $(7, 2)$

Abstract

A hypergraph has property $(7, 2)$ if every subfamily of at most seven edges has a transversal of at most two points. We prove that every finite k -uniform hypergraph with property $(7, 2)$ has transversal number at most $\lceil 6k/7 \rceil$, for every $k \geq 2$. This improves the asymptotic coefficient $7/8$ of Fon-Der-Flaass, Kostochka and Woodall to $6/7$ and includes the rank $k = 7$, not covered by their bound; the conjectured asymptotic value is $3k/4$. The proof refines their avoidance method. Its main new tool is a static allocation of four avoidance requests which, together with a symmetric allocation, closes every “good triple” of edges whose pairwise intersections have at most $T/2$ points, except those with exactly two intersections of medium size (between $T - \lceil k/2 \rceil$ and $T/2$); here $T = \lceil 6k/7 \rceil$. Gap information about pairwise intersection sizes, bootstrapped in three stages, handles the remaining triples. We also explain why $6/7$ appears to be the natural limit of these constructions: three independent constraints of the argument coincide exactly there.

1 Introduction

A *transversal* of a hypergraph is a set of points meeting every edge, and the *transversal number* $\tau(\mathcal{H})$ is the least size of a transversal. Following Erdős, Hajnal and Tuza [4], a hypergraph has *property* (p, t) if every subfamily of at most p edges has a transversal of size at most t . For integers $k \geq 2$ and $p \geq 3$ let $f(k, p)$ denote the largest transversal number of a finite k -uniform hypergraph with property $(p, 2)$. (Using “at most p edges” avoids vacuous exceptions for families with fewer than p edges; for $p = 7$ this is also the convention of the proof in [5].)

Erdős, Fon-Der-Flaass, Kostochka and Tuza [3] determined $f(k, p)$ for $3 \leq p \leq 6$; in particular $f(k, 6) = k$. For $p = 7$ the behaviour changes. Fon-Der-Flaass, Kostochka and Woodall [5] proved

$$f(k, 7) \leq \lceil 7k/8 \rceil \quad (k \geq 8),$$

and constructed $4m$ -uniform families with property $(7, 2)$ and transversal number $3m + 1$ (for $m \geq 10$, and by their remark for $m \geq 4$). Whether $f(k, 7) = (3/4 + o(1))k$ is Problem 644 in Bloom’s list of Erdős problems [1]; it remains open. Kostochka [6] and Bucić, Korándi and Sudakov [2] studied the general parameters for large p ; their results do not apply at $p = 7$.

We improve the coefficient $7/8$ to $6/7$.

Theorem 1.1. *Let $k \geq 2$. Every finite k -uniform hypergraph $\mathcal{H}$ with property $(7, 2)$ satisfies*

$$\tau(\mathcal{H}) \leq \lceil 6k/7 \rceil.$$

The same bound holds for finite families of nonempty sets of size at most k . In particular $f(k, 7) \leq \lceil 6k/7 \rceil$ for all $k \geq 2$, and $\limsup_{k \rightarrow \infty} f(k, 7)/k \leq 6/7$.

For $2 \leq k \leq 6$ we have $\lceil 6k/7 \rceil = k$, and the theorem follows from the upper bound $f(k, 6) \leq k$ of [3] (see also [5, Section 1]). Indeed, property (7, 2) in our sense implies property (6, 2) in either convention; and if $f(k, 6) \leq k$ is read in the convention of exactly six edges, families with at most six edges are covered directly, since they have a transversal of at most $2 \leq k$ points by property (7, 2) itself. The content of Theorem 1.1 is therefore the range $k \geq 7$. The rank $k = 7$ needs one additional construction (Proposition 5.3); for $k \geq 8$ the general argument suffices. The bound $\lceil 6k/7 \rceil$ is smaller than $\lceil 7k/8 \rceil$ for every $k \geq 49$ and for many smaller k ; the two coincide for 21 values of k between 8 and 48. For $k = 7$ the previously available bound was $f(7, 7) \leq f(7, 6) = 7$.

The method

The proof refines the avoidance method of [5]. If $\tau(\mathcal{H}) > T$, then every set of at most T points is disjoint from some edge; we call such a set a *request* and such an edge a *response*. Three edges with empty common intersection form a *good triple*; its three pairwise intersections are its *pair cells*. Four further requests, chosen so that every two-point transversal of the edges obtained so far lies inside some request, produce a subfamily of at most seven edges without a two-point transversal, contradicting property (7, 2). We say that the triple *closes*. Fon-Der-Flaass, Kostochka and Woodall used exactly this scheme, together with case distinctions on pairwise intersection sizes and a final argument based on a largest “small” intersection. We inherit that framework. What is new is the set of closing constructions and the way they are combined.

Put $T = \lceil 6k/7 \rceil$ and $h = k - T = \lfloor k/7 \rfloor$. Call a pair cell *small* if its size is at most $\gamma = T - \lceil k/2 \rceil$, and *medium* if its size lies in $(\gamma, T/2]$. Normalized by k , the thresholds are $5/14$ and $3/7$. The key new ingredient is the following static observation (Corollary 3.2): *a good triple closes by four requests fixed in advance, with no adaptivity, whenever one pair cell has size at most $T/2$, the other two are small, and those two have total size at most $3T - 2k$* . Together with a symmetric static allocation, this settles every triple whose cells are at most $T/2$, except those with exactly two medium cells (Section 5). The two static allocations do not close all triples with two medium cells, nor all triples containing a cell in $(T/2, k/2]$; we treat these with adaptive requests and with information about which intersection sizes occur. That information is produced in three stages. The first stage uses no prior information and excludes intersection sizes just above $T/2$. The second uses the first to exclude an interval of small sizes. The third uses the second to exclude the rest of $(T/2, k/2]$ (Section 6). The second stage and the finishing argument have the same structure: static closure unless two cells are medium, and a gap lemma in that case.

The coefficient $6/7$ is not an artefact of rounding. Three independent constraints of the argument meet exactly at $T = 6k/7$: the load inequality $3k + a \leq 4T$ of the static allocation at $a = T/2$; the requirement that a request built from a pair meeting in q points forces small traces, which needs $q \geq 3k - 3T$; and the budget $2k + M + z \leq 3T$ of the near-core construction at $M = T/2$, $z = k - T$. We explain this in Section 8. It suggests that improving the coefficient requires a new type of construction rather than a retuning of parameters.

Organization

Section 2 sets up requests, labels, integral allocations and good triples. Section 3 proves the two static closures. Section 4 states the adaptive closures; their proofs are in Appendix A. Section 5 proves that it suffices to exclude pair intersections in $(T/2, k/2]$, and Section 6 excludes them. Section 7 completes the proof, and Section 8 discusses the limits of the method. Every step is a hand argument with exact integer quantities; no computation is used in the proof. Table 1 lists, for each step of the global argument, the lemma that closes each of its cases.

Step	Lemma used in each case
Local finisher, Proposition 5.1	Corollary 3.2; Lemma 3.4; Lemma 4.6; Lemma 4.7
Small maximum, Theorem 5.2	Corollary 3.2 (twice); Lemma 4.8
Rank seven, Proposition 5.3	explicit requests; Corollary 3.2
First gap, Proposition 6.2	Lemma 4.2; Lemma 4.1; Lemma 3.4
Middle gap, Proposition 6.3	Corollary 3.2; Lemma 4.6
Upper gap, Proposition 6.4	Lemma 4.3; Lemma 4.4; Lemma 4.5

Table 1: Case structure of the proof. Each case is closed by one lemma.

2 The framework

Throughout Sections 2–4, $\mathcal{H}$ is a finite k -uniform hypergraph and T is an integer with $0 \leq T \leq k$ and $\tau(\mathcal{H}) > T$. A *request* is a set of at most T points. Since $\tau(\mathcal{H}) > T$, every request is disjoint from some edge of $\mathcal{H}$, a *response* to it. A subfamily is *bad* if it has no transversal of size at most two. A bad subfamily of at most seven edges contradicts property $(7, 2)$.

2.1 Covering by requests

Lemma 2.1 (Request-cover principle). *Let $A_1, \dots, A_t$ be edges and $D_1, \dots, D_r$ requests, $t + r \leq 7$. If every transversal of $\{A_1, \dots, A_t\}$ of size at most two is contained in some D_j , then $A_1, \dots, A_t$ together with responses to $D_1, \dots, D_r$ form a bad subfamily of at most seven edges.*

Proof. A transversal of size at most two of the whole subfamily meets $A_1, \dots, A_t$, so it lies in some D_j and misses the response to D_j . Repeated edges only decrease the number of distinct edges. $\square$

Requests may be chosen adaptively: a request may depend on earlier responses. A transversal of size one is contained in a transversal of size two whenever there are at least two points, so we only discuss pairs of points. A pair $\{u, v\}$ of points is a *candidate pair* of a family of edges if it meets every edge of the family.

Labels. Given requests $R_1, \dots, R_r$, the *label* of a point is the set of indices j with the point in R_j . A pair $\{u, v\}$ lies in a common request exactly when the labels of u and v intersect. We often partition the points into *cells* and prescribe the labels allowed in each cell; then two cells whose points form candidate pairs must receive labels that pairwise intersect.

2.2 Integral allocation

Most constructions place some cells in prescribed requests and then distribute other cells among several allowed requests. The following form of the supply–demand theorem guarantees integral distributions.

Lemma 2.2 (Integral allocation). *Let $W_1, \dots, W_n$ be disjoint finite sets. Suppose each W_i has a set $N_i \subseteq \{1, \dots, r\}$ of allowed indices, and each index j has an integer capacity $c_j \geq 0$. The points of $\bigcup W_i$ can be assigned so that every point of W_i receives one index of N_i and index j receives at most c_j points if and only if*

$$\sum_{i \in I} |W_i| \leq \sum_{j \in N(I)} c_j \quad \text{for every } I \subseteq \{1, \dots, n\}, \quad N(I) = \bigcup_{i \in I} N_i.$$

Proof. Necessity is clear. For sufficiency, form the network with arcs from a source to W_i of capacity $|W_i|$, arcs of infinite capacity from W_i to each $j \in N_i$, and arcs from j to a sink of capacity c_j . Consider a cut of finite capacity and let I be the set of indices i such that W_i lies on the source side. No arc of infinite capacity is cut, so $N(I)$ lies on the source side as well, and the capacity of the cut is at least $\sum_{i \notin I} |W_i| + \sum_{j \in N(I)} c_j$, which by hypothesis is at least $\sum_i |W_i|$. By the max-flow min-cut theorem and the integrality of maximum flows with integer capacities, there is an integral flow saturating every source arc; it gives the assignment. $\square$

When $n = 1$ the condition is $|W_1| \leq \sum_{j \in N_1} c_j$. Adding a point to a request never destroys a covering, so a construction may always put a cell into more requests than necessary, as long as the sizes stay at most T .

2.3 Good triples

A triple E, F, G of edges is *good* if $E \cap F \cap G = \emptyset$. Throughout we write

$$X = E \cap F, \quad Y = E \cap G, \quad Z = F \cap G, \quad x = |X|, \quad y = |Y|, \quad z = |Z|, \quad S = x + y + z,$$

and $P_E = E \setminus (F \cup G)$, $P_F = F \setminus (E \cup G)$, $P_G = G \setminus (E \cup F)$, of sizes $k - x - y$, $k - x - z$, $k - y - z$. Because the triple is good, the six cells X, Y, Z, P_E, P_F, P_G are disjoint. A pair meets all three edges exactly when it lies in two different pair cells, or in a pair cell and the opposite private part; so the candidate pairs are

$$X \times Y, X \times Z, Y \times Z, \quad X \times P_G, Y \times P_F, Z \times P_E. \quad (2.1)$$

Points outside $E \cup F \cup G$ belong to no candidate pair.

A triple *closes at budget T* if four further requests of size at most T , chosen possibly adaptively, force a bad subfamily of at most seven edges. Since only sizes of cells enter the constructions, the statement that a triple with given pair sizes closes does not depend on the names of its edges; we permute E, F, G freely and say which cell plays which role.

After a fourth edge K is obtained, candidate pairs of $\{E, F, G, K\}$ are described as follows. Label each point by the set of edges among E, F, G, K containing it.

Lemma 2.3 (Four edges). *Let E, F, G be a good triple and K an edge. The candidate pairs of $\{E, F, G, K\}$ are exactly the pairs in*

$$\begin{aligned} & (X \cap K) \times G, \quad (Y \cap K) \times F, \quad (Z \cap K) \times E, \\ & (X \setminus K) \times (K \cap P_G), \quad (Y \setminus K) \times (K \cap P_F), \quad (Z \setminus K) \times (K \cap P_E). \end{aligned}$$

Proof. A pair meets the four edges exactly when the union of the labels of its points is $\{E, F, G, K\}$. No label contains $\{E, F, G\}$. If one label has three elements, it is $\{E, F, K\}$, $\{E, G, K\}$ or $\{F, G, K\}$ and the other point must lie in the missing edge; this gives the first row. If both labels have at most two elements, they are complementary two-element sets, and one of them contains K ; this gives the second row. A label with at most one element can only be completed by a three-element label. $\square$

The sets $X \cap K$, $Y \cap K$, $Z \cap K$ are the *triple cells* of the four edges. A response might coincide with an earlier edge (for instance when its traces on the earlier edges are not forced to be smaller than k). This is harmless: Lemma 2.1 counts distinct edges, and Lemma 2.3 holds also when K is one of E, F, G . Most adaptive constructions first request an edge K avoiding all or most of the pair cells, so that few triple cells survive.

2.4 Intersection gaps

For integers $0 \leq \ell \leq u < k$ we write $\text{Gap}(\ell, u)$ for the statement that no two distinct edges of $\mathcal{H}$ meet in a number of points in $\{\ell + 1, \dots, u\}$; equivalently, $|A \cap B| \leq u$ implies $|A \cap B| \leq \ell$ for distinct edges A, B . Because $u < k$, an edge meeting a given edge in at most u points is distinct from it, so $\text{Gap}(\ell, u)$ applies to every trace of a response that we bound by u . We also use the *dichotomy with threshold m* : every two distinct edges meet in at most m or in more than $k/2$ points. Two disjoint subsets of one edge cannot both have more than $k/2$ points.

2.5 Seed requests

Lemma 2.4 (Balanced request). *If two edges E, F meet in $M \leq T$ points, a response G to a suitable request of size T forms a good triple with them and satisfies*

$$|E \cap G|, |F \cap G| \leq k - \lfloor (T + M)/2 \rfloor.$$

Proof. Request $E \cap F$ together with $\lfloor (T - M)/2 \rfloor$ points of $E \setminus F$ and $\lceil (T - M)/2 \rceil$ points of $F \setminus E$; these fit since $T \leq k$. The response avoids $E \cap F$, and each of its traces is at most $k - M - \lfloor (T - M)/2 \rfloor = k - \lfloor (T + M)/2 \rfloor$. $\square$

Lemma 2.5 (Capped request). *Suppose two edges meet in q points and $K \geq 0$ is an integer with $0 \leq 2k - T - q \leq 2K$ and $K \leq k - q$. Then a response to a suitable request of size T forms a good triple with the two edges, with both new pair intersections at most K .*

Proof. Choose integers $u, v \in [0, K]$ with $u + v = 2k - T - q$. Keep u points of the first private part and v of the second (possible as $K \leq k - q$) and request all other points of the union of the two edges, which has $2k - q - u - v = T$ points and contains their intersection. $\square$

3 Static closures

A static closure uses four requests fixed in advance. By Lemma 2.1 and (2.1), four requests close a good triple when the three pair cells receive pairwise intersecting labels and each private part receives labels meeting every label of the opposite pair cell.

Lemma 3.1 (Hall allocation). *Let a good triple have pair sizes $a = |X|$, $b = |Y|$, $c = |Z|$. It closes at budget T by a static allocation if*

$$T - a \geq 0, \quad T - a - b \geq 0, \quad T - a - c \geq 0, \quad T - b - c \geq 0, \quad (3.1)$$

$$k + 2b \leq 2T, \quad k + 2c \leq 2T, \quad k + 3a \leq 3T, \quad 2k + b + c \leq 3T, \quad (3.2)$$

$$2k + 2a + b \leq 4T, \quad 2k + 2a + c \leq 4T, \quad 3k + a \leq 4T.$$

Proof. Give the points of X, Y, Z the labels $\{1, 2, 3\}, \{2, 4\}, \{3, 4\}$; these pairwise intersect. Every point of P_G will receive one of $\{1\}, \{2\}, \{3\}$, every point of P_F one of $\{2\}, \{4\}$, and every point of P_E one of $\{3\}, \{4\}$. Each of these meets the label of the opposite pair cell, so all candidate pairs (2.1) are covered. After the pair cells are placed, the remaining capacities of the four requests are $T - a$, $T - a - b$, $T - a - c$, $T - b - c$, which are nonnegative by (3.1). The three private parts have sizes $k - a - b$, $k - a - c$, $k - b - c$ and allowed index sets $\{3, 4\}, \{2, 4\}, \{1, 2, 3\}$. The seven conditions of Lemma 2.2 are, for $P_E, P_F, P_G, P_E \cup P_F, P_E \cup P_G, P_F \cup P_G$ and all three, exactly the seven inequalities (3.2); for example, for P_G the condition is $k - b - c \leq (T - a) + (T - a - b) + (T - a - c)$, that is $k + 3a \leq 3T$, and for all three it is $3k - 2a - 2b - 2c \leq 4T - 3a - 2b - 2c$, that is $3k + a \leq 4T$. $\square$

The next corollary is the form in which the Hall allocation is used.

Corollary 3.2. *Let $T \geq 6k/7$. A good triple closes at budget T if one pair cell has size $a \leq T/2$ and the other two have sizes $b, c \leq T - k/2$ with $b + c \leq 3T - 2k$.*

Proof. We check Lemma 3.1. We have $T - a \geq T/2$, $T - a - b \geq T - T/2 - (T - k/2) = (k - T)/2 \geq 0$, likewise $T - a - c \geq 0$, and $T - b - c \geq T - (3T - 2k) = 2(k - T) \geq 0$. Next $k + 2b, k + 2c \leq 2T$ and $2k + b + c \leq 3T$ hold by hypothesis. Finally, using $T \geq 6k/7$,

$$\begin{aligned} k + 3a &\leq k + \frac{3}{2}T \leq 3T, & 2k + 2a + b &\leq 2k + T + (T - \frac{k}{2}) = \frac{3}{2}k + 2T \leq 4T, \\ 3k + a &\leq 3k + \frac{T}{2} \leq 4T. \end{aligned}$$

The last inequality is equivalent to $7T \geq 6k$. □

The second static closure treats triples whose three cells are all of roughly the same size. It rests on an exact allocation lemma.

Lemma 3.3 (Boxed triangle). *Let $M_1, M_2, M_3 \geq 0$ and L_1, L_2, L_3 be integers with $L_i \leq M_j + M_l$ whenever $\{i, j, l\} = \{1, 2, 3\}$, and let $t \geq 0$ be an integer. There are integers $0 \leq a_i \leq M_i$ with $a_j + a_l \geq L_i$ for all $\{i, j, l\} = \{1, 2, 3\}$ and $a_1 + a_2 + a_3 \leq t$ if and only if*

$$t \geq \max\left\{L_1, L_2, L_3, L_1 + L_2 - M_3, L_1 + L_3 - M_2, L_2 + L_3 - M_1, \frac{1}{2}(L_1 + L_2 + L_3)\right\}.$$

Proof. If such a_i exist with $\sigma = a_1 + a_2 + a_3 \leq t$, then $\sigma \geq L_i$ because $a_i \geq 0$; adding $a_2 + a_3 \geq L_1$ and $a_1 + a_3 \geq L_2$ gives $\sigma + a_3 \geq L_1 + L_2$, so $\sigma \geq L_1 + L_2 - M_3$, and similarly for the other two pairs; and adding all three constraints gives $2\sigma \geq L_1 + L_2 + L_3$.

Conversely, suppose t satisfies the displayed inequality. Every displayed quantity is at most $M_1 + M_2 + M_3$: for instance $L_1 + L_2 - M_3 \leq (M_2 + M_3) + (M_1 + M_3) - M_3$ and $L_1 + L_2 + L_3 \leq 2(M_1 + M_2 + M_3)$. So $t' = \min(t, M_1 + M_2 + M_3)$ also satisfies it, and we look for a_i with $a_1 + a_2 + a_3 = t'$. Then the constraints become $0 \leq a_i \leq \min(M_i, t' - L_i)$, and integers with sum t' exist if and only if each upper bound is nonnegative and the three upper bounds sum to at least t' . The first requirement is $t' \geq L_i$. Expanding the sum of three minima as the minimum of eight sums, the second requirement is $M_1 + M_2 + M_3 \geq t'$, $M_j + M_l + t' - L_i \geq t'$ (the host conditions), $M_i + 2t' - L_j - L_l \geq t'$, and $3t' - L_1 - L_2 - L_3 \geq t'$. All hold. □

Lemma 3.4 (Symmetric allocation). *Let a good triple have pair sizes $m \geq y \geq z$ (in any order of the cells) and $S = m + y + z$. It closes at budget T by a static allocation if*

$$T \geq k + m - y - z, \quad 2T \geq k + m, \quad 3T \geq 2k + m + y - z, \quad 5T \geq 3k + S.$$

Proof. Call the pair cells M, Y, Z (sizes m, y, z) and the opposite private parts P_M, P_Y, P_Z . Split $M = M_1 \sqcup M_2$, $Y = Y_1 \sqcup Y_2$, $Z = Z_1 \sqcup Z_2$ with $|M_1| = a_1$, $|Y_1| = a_2$, $|Z_1| = a_3$, and use the requests

$$\begin{aligned} R_0 &= M_1 \cup Y_1 \cup Z_1, & R_M &= M \cup P_M \cup Y_2 \cup Z_2, \\ R_Y &= Y \cup P_Y \cup M_2 \cup Z_2, & R_Z &= Z \cup P_Z \cup M_2 \cup Y_2. \end{aligned}$$

The parts M_1, Y_1, Z_1 have labels $\{0, M\}, \{0, Y\}, \{0, Z\}$, the parts M_2, Y_2, Z_2 all have label $\{M, Y, Z\}$, and P_M, P_Y, P_Z have labels $\{M\}, \{Y\}, \{Z\}$. Any two labels of different pair cells intersect, and each private label meets both labels of its opposite pair cell, so all candidate pairs are covered. The sizes of the four requests are

$$a_1 + a_2 + a_3, \quad k + m - a_2 - a_3, \quad k + y - a_1 - a_3, \quad k + z - a_1 - a_2.$$

Hence it suffices to find integers $0 \leq a_1 \leq m$, $0 \leq a_2 \leq y$, $0 \leq a_3 \leq z$ with $a_2 + a_3 \geq k + m - T$, $a_1 + a_3 \geq k + y - T$, $a_1 + a_2 \geq k + z - T$ and $a_1 + a_2 + a_3 \leq T$. Apply Lemma 3.3 with $(M_1, M_2, M_3) = (m, y, z)$, $(L_1, L_2, L_3) = (k + m - T, k + y - T, k + z - T)$ and $t = T$. The host conditions reduce to $T \geq k + m - y - z$ (the other two are weaker since $m \geq y \geq z$). The largest single L_i gives $2T \geq k + m$; the largest of the three two-term expressions is $L_1 + L_2 - M_3$, which gives $3T \geq 2k + m + y - z$; and the last expression gives $5T \geq 3k + S$. $\square$

4 Adaptive closures

The following lemmas each describe one adaptive construction: a first request produces a fourth edge K , and three further requests, depending on K , cover all candidate pairs of $\{E, F, G, K\}$ given by Lemma 2.3. The hypotheses are exact integer conditions. Proofs are given in Appendix A. We use the notation of Section 2 and assume $T \leq k$ throughout; each lemma may be applied to any assignment of the names E, F, G to the edges of a good triple, which permutes x, y, z .

4.1 Without gap information

The first lemma requests an edge avoiding two pair cells and as much of the third as possible; at most one triple cell survives. The next two avoid all three pair cells.

Lemma 4.1 (One surviving cell). *A good triple closes at budget T if*

$$\begin{aligned} T \geq x + y, \quad 2T \geq k + 2x, \quad 2T \geq k + 2y, \quad T \geq k + x - y - z, \quad T \geq k - x + y - z, \\ 3T \geq 3k - S, \quad 5T \geq 3k + S, \quad 3T \geq k + x + y + 2z, \quad 4T \geq 2k + 3z. \end{aligned}$$

Lemma 4.2 (First asymmetric split). *A good triple closes at budget T if*

$$T \geq S, \quad 2T \geq k + 2y, \quad 2T \geq k + 2x - y + z, \quad 3T \geq k + 2x + y + 3z.$$

Lemma 4.3 (Second asymmetric split). *A good triple closes at budget T if*

$$T \geq S, \quad 3T \geq k + 3x, \quad T \geq k - x + z, \quad 3T \geq k + 2x + 3y + z.$$

4.2 With gap information

In the next three lemmas $\text{Gap}(\ell, u)$ holds for integers $0 \leq \ell \leq u < k$. The first request is designed so that some traces of K are at most u , and hence at most ℓ .

Lemma 4.4 (Two forced traces). *Assume $\text{Gap}(\ell, u)$ and put $p = (k - x - y - u)_+$ and $t = (k - x - z - u)_+$. A good triple closes at budget T if*

$$S + p + t \leq T, \quad k + 3x + p + t \leq 3T, \quad y + z + 2\ell \leq T.$$

Lemma 4.5 (Forced traces and one core). *Assume $\text{Gap}(\ell, u)$ and put $\Delta = (S - T)_+$. A good triple closes at budget T if*

$$\begin{aligned} x + y \leq T, \quad k - x - y \leq u, \quad k - x - z + \Delta \leq u, \\ 2x + 2\Delta + k - y - z \leq 2T, \quad y + z + 2\ell \leq T, \quad 2k + x - y + \ell + \Delta \leq 3T. \end{aligned}$$

Lemma 4.6 (Two cores). *Assume $\text{Gap}(\ell, u)$ and put $g = (k - y - u)_+$. A good triple closes at budget T if*

$$\begin{aligned} g \leq \min(x, z), \quad y + 2g \leq T, \quad x + \ell \leq T, \quad z + \ell \leq T, \\ k + 2y \leq 2T, \quad 3k \leq 4T, \quad 3k + S \leq 5T. \end{aligned}$$

The last two lemmas use the dichotomy with threshold m , where $2m \leq k$.

Lemma 4.7 (Near core). *Assume the dichotomy with threshold m and put $\Delta = (S - T)_+$. A good triple closes at budget T if*

$$\begin{aligned} y + z \leq T, \quad m + z \leq T, \quad x + m \leq T, \quad k + y - z \leq T, \\ 2k + x - 2z \leq 2T, \quad 2k + x + m - z \leq 3T, \end{aligned}$$

and either

$$k - x + y + \Delta \leq T, \quad 2k - 2x + z + \Delta \leq 2T, \quad 2k - z + \Delta \leq 2T, \quad 3k - x - y + \Delta \leq 3T,$$

or $S \leq T$, $k - x + y \leq T$ and $2k - 2x + z \leq 2T$.

Lemma 4.8 (Two large opposite cells). *Assume the dichotomy with threshold m , where $4m \leq k$. Let E, F, G, H be edges with all four triple intersections empty, such that $X = E \cap F$ and $B = G \cap H$ have sizes $x, b > k/2$ and the other four pairwise intersections have at most m points. Three further requests force a bad subfamily of at most seven edges if*

$$T \geq \lceil k/2 \rceil + 2m, \quad T \geq x + m, \quad 3T \geq 2x + b + \lceil k/2 \rceil + 2m.$$

5 The finishing theorem

In this section we show that it is enough to exclude pair intersections of size in $(T/2, k/2]$. Write

$$h = k - T, \quad \gamma = T - \lceil k/2 \rceil.$$

An integer w satisfies $w \leq \gamma$ if and only if $w \leq T - k/2$. We call a pair cell *small* if its size is at most γ and *medium* if its size lies in $(\gamma, T/2]$.

Proposition 5.1 (Local finisher). *Let $6k/7 \leq T \leq k$ and let $M \leq T/2$ be an integer. Assume the dichotomy with threshold M . Then every good triple with pair sizes $M \geq y \geq z$ closes at budget T .*

Proof. We say that the triple has *two medium cells* if $y > \gamma \geq z$. (Its cell of size M is then medium as well.) There are four cases.

Case 1: not two medium cells, and $y + z \leq 3T - 2k$. If $z > \gamma$ then $y + z \geq 2\gamma + 2 \geq 2T - k + 1 > 3T - 2k$, so here $z \leq y \leq \gamma$. Corollary 3.2 applies with $a = M$, $b = y$, $c = z$.

Case 2: not two medium cells, and $y + z > 3T - 2k$. Either $y \leq \gamma$, or $z > \gamma$. We check Lemma 3.4 with $m = M$. First, $M \leq T/2 \leq 4T - 3k$ because $7T \geq 6k$, so $k + M - T \leq 3T - 2k < y + z$. Next $2T \geq k + T/2 \geq k + M$. For the third and fourth conditions we show

$$M + y - z < k - T/2 \quad \text{and} \quad S \leq 3T/2.$$

If $z > \gamma$, then $y - z < M - (T - k/2)$, so $M + y - z < 2M - T + k/2 \leq k/2 \leq k - T/2$; and $S \leq 3M \leq 3T/2$. If $y \leq \gamma$, then $y - z = 2y - (y + z) < 2(T - k/2) - (3T - 2k) = k - T$, so $M + y - z < k - T/2$; and $S \leq T/2 + 2(T - k/2) \leq 3T/2$. Consequently $2k + M + y - z < 3k - T/2 \leq 3T$ and $3k + S \leq 3k + 3T/2 \leq 5T$, both because $7T \geq 6k$.

Case 3: two medium cells and $z > h$. Apply Lemma 4.6 with the cell of size z in the role of Y and the cells of sizes M, y in the roles of X, Z ; that is, with (x, y, z) of the lemma equal to (M, z, y) , and with $\text{Gap}(M, \lfloor k/2 \rfloor)$, which is implied by the dichotomy (note $0 \leq M \leq \lfloor k/2 \rfloor < k$). Here $g = (k - z - \lfloor k/2 \rfloor)_+ = \lceil k/2 \rceil - z$, as $z \leq \gamma \leq \lfloor k/2 \rfloor$. Since $M + z \geq y + z > (T - k/2) + (k - T) = k/2$, both M and y are at least $\lceil k/2 \rceil - z = g$. The first request has size $z + 2g = \max\{z, k - z + (k - 2 \lfloor k/2 \rfloor)\} \leq T$, because $z \geq h + 1$. The remaining conditions are

$$M + M \leq T, \quad y + M \leq T, \quad k + 2z \leq k + 2\gamma \leq 2T, \quad 3k \leq 4T,$$

$$\text{and } 3k + S \leq 3k + 2M + \gamma \leq 3k + T + T - k/2 = \frac{5}{2}k + 2T \leq 5T.$$

Case 4: two medium cells and $z \leq h$. Apply Lemma 4.7 with $m = M$ and (x, y, z) of the lemma equal to (M, z, y) . With $c = T - k/2$ we have $y > c$, $z \leq h$ and $h \leq c$. The common conditions are

$$\begin{aligned} z + y \leq h + T/2 = k - T/2 \leq T, \quad M + y \leq 2M \leq T, \quad M + M \leq T, \\ k + z - y < k + h - c = \frac{5}{2}k - 2T \leq T, \quad 2k + M - 2y < 2k + \frac{T}{2} - 2c = 3k - \frac{3}{2}T \leq 2T, \\ 2k + 2M - y < 2k + T - c = \frac{5}{2}k \leq 3T. \end{aligned}$$

If $S \leq T$ we use the second alternative: $k - M + z \leq k - y + z < T$ and $2k - 2M + y \leq 2k - M < 2k - c = \frac{5}{2}k - T \leq 2T$. If $S > T$ then $\Delta = S - T$ and the four remaining conditions become

$$\begin{aligned} k + y + 2z \leq k + \frac{T}{2} + 2h = 3k - \frac{3}{2}T \leq 2T, \quad 2k - M + 2y + z \leq 2k + y + z \leq 3k - \frac{T}{2} \leq 3T, \\ 2k + M + z \leq 2k + \frac{T}{2} + h = 3k - \frac{T}{2} \leq 3T, \quad 3k + y \leq 3k + \frac{T}{2} \leq 4T, \end{aligned}$$

where we used $y \leq M \leq T/2$ and $z \leq h$; each final inequality is equivalent to $T \geq 6k/7$ or weaker.

The four cases cover all triples. $\square$

Theorem 5.2 (Finishing theorem). *Let $k \geq 7$ and $T = \lceil 6k/7 \rceil$. If every two distinct edges of a finite k -uniform hypergraph $\mathcal{H}$ with property (7, 2) meet in at most $T/2$ or in more than $k/2$ points, then $\tau(\mathcal{H}) \leq T$.*

Proof. Suppose $\tau(\mathcal{H}) > T$. An edge avoiding a T -subset of another edge meets it in at most $h = k - T \leq k/2$ points, so the largest size M of a pair intersection that is at most $k/2$ exists. By hypothesis $M \leq T/2$, and by the choice of M the dichotomy with threshold M holds. Write $k = 7h + s$ with $0 \leq s \leq 6$; then $h = \lfloor k/7 \rfloor \geq 1$.

Case $M \geq h + 1$. (For $k = 7$ this is the case $M \in \{2, 3\}$.) Apply Lemma 2.4 to a pair attaining M . As $T + M \geq k + 1$, the new pair intersections are at most $k - \lceil k/2 \rceil = \lfloor k/2 \rfloor$, hence at most M . The resulting good triple closes by Proposition 5.1, a contradiction.

Case $M \leq h$ and $k = 7$. Then $h = 1$, so every two edges meet in at most one or at least four points, and Proposition 5.3 below gives a contradiction.

Case $M \leq h$ and $k \geq 8$. Then $h + s \geq 2$. We record $\gamma \geq h$, $2h \leq 3T - 2k$ and $4h \leq k$, which follow from $k \geq 7h$. Apply Lemma 2.4 to edges E, G with $|E \cap G| = M$, obtaining F . If both $|E \cap F|$ and $|F \cap G|$ are at most $k/2$, they are at most M , and the triple closes by Corollary 3.2, as all three cells are at most $h \leq \gamma$ and the two cells other than $E \cap G$ sum to at most $2h \leq 3T - 2k$. Otherwise exactly one of them exceeds $k/2$, since they are disjoint subsets of F ; by symmetry

$$x = |E \cap F| > k/2, \quad |E \cap G| = M, \quad |F \cap G| \leq M,$$

and $x \leq k - \lfloor (T + M)/2 \rfloor \leq k - (T + M - 1)/2$. Then

$$x + 2M \leq k - \frac{T}{2} + \frac{3}{2}M + \frac{1}{2} \leq \frac{k}{2} + 2h + \frac{1}{2} \leq T,$$

using $T = k - h$, $M \leq h$ and $k \geq 6h + 1$. Request $X \cup Y \cup Z$, which has $x + |Y| + |Z| \leq x + 2M \leq T$ points, together with $T - x - |Y| - |Z|$ private points of G (there are enough, as $|P_G| = k - |Y| - |Z|$), and let H be a response. All triple intersections of E, F, G, H are empty. Since H avoids X , its traces on E and F have at most $k - x < k/2$ points, hence at most M . Its trace on G has $b \leq k - T + x$ points.

If $b \leq k/2$, then $b \leq M$ and E, G, H is a good triple (as H avoids $Y = E \cap G$) with all cells at most $M \leq h$; it closes by Corollary 3.2 exactly as above. We discard F : the resulting bad subfamily consists of E, G, H and four responses.

If $b > k/2$, apply Lemma 4.8 to E, F, G, H with $m = M$. Its conditions hold:

$$\begin{aligned} \lfloor k/2 \rfloor + 2M &\leq \frac{k+1}{2} + 2h \leq k - h = T, & x + M &\leq k - \frac{T}{2} + \frac{M}{2} + \frac{1}{2} \leq \frac{k}{2} + h + \frac{1}{2} \leq T, \\ 2x + b + \lfloor k/2 \rfloor + 2M &\leq 3x + h + \frac{k+1}{2} + 2M \leq \frac{7}{2}k - \frac{3}{2}T + \frac{M}{2} + h + 2 \leq 2k + 3h + 2 \leq 3T, \end{aligned}$$

where the final step is $k \geq 6h + 2$, that is $h + s \geq 2$. The resulting bad subfamily has seven edges. In every case property (7, 2) is violated. $\square$

At $k = 7$ the last inequality fails: for $x = |E \cap F| = 4$, $b = |G \cap H| = 5$ and $M = 1$, the third condition of Lemma 4.8 reads $18 \geq 19$. A differently aimed fourth request repairs this.

Proposition 5.3 (Rank seven). *Let $\mathcal{H}$ be a finite 7-uniform hypergraph with property (7, 2) in which every two distinct edges meet in at most one or at least four points. Then $\tau(\mathcal{H}) \leq 6$.*

Proof. Suppose $\tau(\mathcal{H}) > 6$, so requests have at most six points. By hypothesis, a response that misses some point of an earlier edge meets that edge in 0, 1, or at least 4 points; in particular, *if it meets the edge in at most three points, then it meets it in at most one point*. An edge avoiding six points of another edge meets it in at most one point.

Case 1: no two edges meet in exactly one point. Then some edges E_1, E_2 are disjoint: an edge avoiding six points of another meets it in at most one point, hence in none. Request three points of each. The response G meets each E_i within four points, hence in 0 or 4 points, and not in 4 points of both since $|G| = 7$. If G is disjoint from both, the edges E_1, E_2, G are pairwise disjoint and form a bad subfamily. Otherwise, say, $Q = G \cap E_1$ has four points and $G \cap E_2 = \emptyset$. Every candidate pair of $\{E_1, E_2, G\}$ has a point of E_2 , and its other point lies in $E_1 \cap G = Q$. Partition E_2 into sets $S_1, \dots, S_4$ of sizes 2, 2, 2, 1; the four requests $Q \cup S_i$ cover $Q \times E_2$.

Case 2: two edges E, F meet in one point p . Request p , three points of $E \setminus F$ and two points of $F \setminus E$. The response G avoids p , so E, F, G is a good triple; $|G \cap E| \leq 1$ since G meets E within three points, and $|G \cap F| \in \{0, 1, 4\}$. If $|G \cap F| \leq 1$, all three pair cells have at most one point and the triple closes by Corollary 3.2 ($T = 6 = 6k/7$, and $a \leq 3$, $b, c \leq 1 \leq T - k/2$, $b + c \leq 2 \leq 3T - 2k$).

It remains to treat $|G \cap F| = 4$. Rename $A = F$, $B = G$, $C = E$. Then $X = A \cap B$ has four points, $A \cap C = \{y\}$ with $y = p$, $Z = B \cap C$ has at most one point, and $A \cap B \cap C = \emptyset$. Put $A' = A \setminus B$ (three points, including y), $B' = B \setminus A$ (three points, including Z) and $P_C = C \setminus (A \cup B)$ (5 or 6 points). Choose $x_1 \in X$, $c_1 \in P_C$ and $b^* \in B'$ with $Z \subseteq \{b^*\}$, and request

$$D = \{x_1\} \cup A' \cup \{b^*, c_1\} \quad (|D| = 6).$$

Let H be a response; it differs from A, B, C . Its traces satisfy: $H \cap A \subseteq X \setminus \{x_1\}$, so $|H \cap A| \leq 1$ and $H \cap A = H \cap X$; $H \cap B \subseteq (X \setminus \{x_1\}) \cup (B' \setminus \{b^*\})$ has at most $1 + 2$ points, so $|H \cap B| \leq 1$;

and $H \cap C \subseteq P_C \setminus \{c_1\}$, so $\eta = |H \cap C| \in \{0, 1, 4, 5\}$. Consequently H meets at most one of X and B' , each in at most one point.

A candidate pair of $\{A, B, C, H\}$ contains a point of X , or consists of $u \in A'$ and $v \in B'$. In the first case the other point lies in C (as $X \cap C = \emptyset$), and in H unless the X -point is in H . In the second case $u \notin H$, so $v \in H \cap B'$; then $v \neq b^*$, so $v \notin C$, and $u \in C$ forces $u = y$. Hence the candidate pairs are

$$\{x\} \times C \text{ if } H \cap X = \{x\}, \quad (X \setminus H) \times (H \cap C), \quad \{y\} \times (H \cap B').$$

Write $X = \{x_1, x_2, x_3, x_4\}$ and, when $\eta = 5$, $H \cap C = \{v_1, \dots, v_5\}$. The following requests, all of size at most six, cover these pairs; together with A, B, C, H there are at most seven edges.

$H \cap X = \{x\}$. Then $H \cap B' = \emptyset$. Let $x' \in X \setminus \{x\}$. If $\eta \leq 1$, choose $S \subseteq C \setminus H$ with $|S| = 2 - \eta$ and use $X \cup (H \cap C) \cup S$ and $\{x\} \cup (C \setminus (H \cup S))$. If $\eta = 4$, use $\{x, x'\} \cup (H \cap C)$, $(X \setminus \{x, x'\}) \cup (H \cap C)$ and $\{x\} \cup (C \setminus H)$. If $\eta = 5$, use $X \cup \{v_1, v_2\}$, $\{x\} \cup (C \setminus \{v_1, v_2\})$ and $(X \setminus \{x\}) \cup \{v_3, v_4, v_5\}$.

$H \cap X = \emptyset$. Put $Y_b = \{y\} \cup (H \cap B')$, of at most two points. If $\eta \leq 1$, use $X \cup (H \cap C)$ and Y_b . If $\eta = 4$, use $\{x_1, x_2\} \cup (H \cap C)$, $\{x_3, x_4\} \cup (H \cap C)$ and Y_b . If $\eta = 5$, use $X \cup \{v_1, v_2\}$, $\{x_1\} \cup Y_b \cup \{v_3, v_4, v_5\}$ and $\{x_2, x_3, x_4, v_3, v_4, v_5\}$.

In each case every candidate pair lies in one of the listed requests, which completes the proof. $\square$

6 Three intersection gaps

In Sections 6 and 7 (up to the final padding step) fix $k \geq 7$ and write

$$h = \lfloor k/7 \rfloor, \quad s = k - 7h, \quad T = k - h = \lceil 6k/7 \rceil,$$

so that $h \geq 1$, $0 \leq s \leq 6$ and $6k/7 \leq T \leq k$. Assume $\tau(\mathcal{H}) > T$ for a finite k -uniform family $\mathcal{H}$ with property (7, 2). Define the integers

$$\alpha = \lfloor T/2 \rfloor, \quad \beta = \lfloor (4T - 2k)/3 \rfloor, \quad \gamma = T - \lceil k/2 \rceil, \quad \delta = 2k - T - 2\beta, \quad \lambda = \delta - 1. \quad (6.1)$$

Normalized by k , these are approximately $3/7, 10/21, 5/14, 4/21, 4/21$. The first stage excludes pair intersections in $(\alpha, \min(\beta, \lfloor k/2 \rfloor)]$. The other two stages, which exclude $[\delta, \gamma]$ and then $(\beta, k/2]$, are needed only when $\beta < \lfloor k/2 \rfloor$; only then are δ and λ used, and only then do we need $\lambda \geq 0$, which Lemma 6.1 guarantees (outside this branch λ can be negative, for instance $\lambda = -1$ for $k = 11$).

Lemma 6.1 (Endpoints). *Put $r = \lfloor (h + 2s)/3 \rfloor$. Then*

$$\alpha = 3h + \lfloor s/2 \rfloor, \quad \beta = 3h + r, \quad \gamma = \lfloor (5h + s)/2 \rfloor, \quad \delta = 2h + s - 2r, \quad \lambda = 2h + s - 2r - 1,$$

and

$$0 \leq \gamma \leq \alpha \leq \beta, \quad \beta \geq 3h, \quad \lambda \leq \gamma - h, \quad 4\lambda \leq T, \quad k - \beta - 1 + \lambda \leq T. \quad (6.2)$$

If $\beta < \lfloor k/2 \rfloor$, then $h < \delta \leq \gamma$, so $0 \leq \lambda < \gamma < k$, and moreover $T + \beta - k \leq \gamma - 1$.

Proof. The formulas follow from $k = 7h + s$ and $T = 6h + s$. Clearly $\gamma \geq 0$, and $\gamma \leq \alpha$ follows from $T \leq k$. As $T \geq 4k/5$ we have $(4T - 2k)/3 \geq T/2$, so $\beta \geq \alpha$; and $\beta \geq 3h$ is clear. For $\lambda \leq \gamma - h = \lfloor (3h + s)/2 \rfloor$ it suffices that $2r + 1 \geq \lfloor (h + s)/2 \rfloor$; since $r \geq \lfloor (h + s)/3 \rfloor$, this follows from $\lceil n/2 \rceil \leq 2 \lfloor n/3 \rfloor + 1$ for integers $n \geq 0$ (write $n = 3a + b$ with $0 \leq b \leq 2$; then $n \leq 4a + 2$). Writing $h + 2s = 3r + e$ with $0 \leq e \leq 2$, we get $4\lambda - T = 2e - 2r - s - 4 \leq 0$ and $k - \beta - 1 + \lambda - T = s - 3r - 2 \leq -h - s \leq 0$.

If $\beta < \lfloor k/2 \rfloor$, then $2\beta < k$ gives $\delta > k - T = h$, and $\delta = \lambda + 1 \leq \gamma - h + 1 \leq \gamma$. Finally $T + \beta - k = \beta - h \leq \lfloor (7h + s)/2 \rfloor - 1 - h = \lfloor (5h + s)/2 \rfloor - 1 = \gamma - 1$. $\square$

Proposition 6.2 (First gap). *No two edges meet in q points with $\alpha < q \leq \min(\beta, \lfloor k/2 \rfloor)$.*

Proof. Apply Lemma 2.5 with cap γ : we have $\gamma \leq \lfloor k/2 \rfloor \leq k - q$, and $2k - T - 2\gamma = 3h + (k - 2\lfloor k/2 \rfloor) \leq \alpha + 1 \leq q \leq 2k - T$. The resulting good triple has cells $x = q$ and $\gamma \geq y \geq z \geq 0$. Put $d = y - z$, $v = y + z$ and $S = x + v$. We use $2y \leq 2\gamma \leq 2T - k$, $3h < x \leq \beta$, and $\beta \leq (4T - 2k)/3 \leq 3T - 2k$ (the latter as $T \geq 4k/5$).

Case 1: $d > x - h$. Lemma 4.2 applies to (x, y, z) . From $v + d = 2y \leq 2T - k$ we get $S \leq x + 2T - k - d < 2T - k + h = T$. Next $2T \geq k + 2y$. Then $k + 2x - y + z = k + 2x - d < k + x + h \leq 2T$, as $x \leq 3T - 2k$. Finally $k + 2x + y + 3z = k + 2x + 2v - d < k + 2x + 4T - 2k - 3(x - h) = T + 2k - x \leq 3T$, as $x \geq 2h$.

Case 2: $d \leq x - h$ and $v \leq 3T - k - 2x$. Lemma 4.1 applies with (x, y, z) of the lemma equal to (y, z, x) . Its nine conditions read

$$\begin{aligned} v \leq 2T - k \leq T, \quad k + 2y \leq 2T, \quad k + 2z \leq 2T, \quad k + d - x \leq k - h = T, \quad k - d - x \leq T, \\ 3k - S \leq 3k - 3h = 3T, \quad 3k + S \leq 2k + x + 2T \leq 5T, \quad k + v + 2x \leq 3T, \quad 2k + 3x \leq 4T, \end{aligned}$$

using $S \geq x \geq 3h$, $d \geq 0$ and $x \geq h$ (for $k - d - x \leq k - x \leq T$), $S \leq x + 2T - k$, $x \leq 3T - 2k$, the case hypothesis and $x \leq \beta$.

Case 3: $v > 3T - k - 2x$. Lemma 3.4 applies with $m = x$, the largest cell. Indeed $k + x - v < 2k + 3x - 3T \leq T$; $k + x \leq 2T$ because $x \leq \beta \leq 2T - k$; $d = 2y - v \leq 2T - k - v < 2x - T$, so $2k + x + d < 2k + 3x - T \leq 3T$; and $3k + S \leq 5T$ as in Case 2. Here $2k + 3x - T \leq 3T$ is again $x \leq \beta$. $\square$

If $\beta \geq \lfloor k/2 \rfloor$, Proposition 6.2 already excludes every pair intersection in $(\alpha, k/2]$. For the next two stages we therefore assume

$$\beta < \lfloor k/2 \rfloor, \tag{6.3}$$

so that Lemma 6.1 gives $h < \delta \leq \gamma$ and $0 \leq \lambda < \gamma$, and Proposition 6.2 gives $\text{Gap}(\alpha, \beta)$.

Proposition 6.3 (Middle gap). *Under (6.3), no two edges meet in q points with $\delta \leq q \leq \gamma$.*

Proof. Apply Lemma 2.5 with cap β : $\beta < \lfloor k/2 \rfloor \leq k - q$ and $0 \leq 2k - T - q \leq 2k - T - \delta = 2\beta$. Both new cells are at most β , hence at most α by $\text{Gap}(\alpha, \beta)$. So the good triple has cells $\delta \leq x \leq \gamma$ and $\alpha \geq y \geq z \geq 0$. This is the structure of the finishing argument: x is small and $y, z \leq T/2$.

Case 1: $x + z \leq k - \beta - 1$. Apply Corollary 3.2 with $a = y$, $b = x$, $c = z$. Here $x \leq \gamma$, $z \leq k - \beta - 1 - \delta = T + \beta - k - 1 \leq \gamma$ by Lemma 6.1, and $x + z \leq k - \beta - 1 \leq k - 3h - 1 < 3T - 2k$.

Case 2: $x + z \geq k - \beta$. Apply Lemma 4.6 with (x, y, z) of the lemma equal to (y, x, z) and $\text{Gap}(\alpha, \beta)$. Then $g = (k - x - \beta)_+ \leq \min(y, z)$ because $z \geq k - \beta - x$ and $y \geq z$. The first request has size $x + 2g = \max(x, 2k - x - 2\beta) \leq T$ because $x \geq \delta$. The remaining conditions are $y + \alpha$, $z + \alpha \leq 2\alpha \leq T$, $k + 2x \leq k + 2\gamma \leq 2T$, $3k \leq 4T$, and $3k + S \leq 3k + \gamma + 2\alpha \leq \frac{5}{2}k + 2T \leq 5T$. $\square$

Proposition 6.4 (Upper gap). *Under (6.3), no two edges meet in q points with $\beta < q \leq \lfloor k/2 \rfloor$.*

Proof. Apply Lemma 2.5 with cap γ , as in Proposition 6.2 (now $q \geq \beta + 1 \geq 3h + 1$). Both new cells are at most γ , hence below δ by Proposition 6.3; that is, $\text{Gap}(\lambda, \gamma)$ holds and the triple has cells

$$\beta < x \leq \lfloor k/2 \rfloor, \quad \lambda \geq y \geq z \geq 0.$$

Put $S = x + y + z$ and $J = 2k - \gamma + 2 \lfloor k/2 \rfloor - 3T = \lfloor (h - s)/2 \rfloor$. We use $\lambda \leq \gamma - h$, $4\lambda \leq T$ and $k - x + y \leq k - \beta - 1 + \lambda \leq T$ from Lemma 6.1.

Case 1: $S \leq T$, and $S \leq 2T - k$ or $z < J$. Apply Lemma 4.3 with (x, y, z) of the lemma equal to (x, z, y) . Its conditions are $S \leq T$, $k + 3x \leq \frac{5}{2}k \leq 3T$, $k - x + y \leq T$, and $k + 2x + 3z + y \leq 3T$. For the last, if $S \leq 2T - k$ then $2x + 3z + y \leq 2S \leq 4T - 2k$ (as $y \geq z$), so $k + 2x + 3z + y \leq 4T - k \leq 3T$; if $z < J$ then $z < h/2$, $y \leq \lambda \leq 2h + s$ and $2x \leq k$ give $k + 2x + 3z + y \leq 2k + \frac{3}{2}h + 2h + s \leq 18h + 3s = 3T$.

Case 2: $2T - k < S \leq T$ and $z \geq J$. Apply Lemma 4.4 with $\text{Gap}(\lambda, \gamma)$. Here

$$S + p + t = \max(S, k - \gamma + z, k - \gamma + y, 2k - 2\gamma - x) \leq T,$$

since $k - \gamma + y \leq k - h = T$ and $2k - 2\gamma - T = 3h + (k - 2 \lfloor k/2 \rfloor) \leq \beta + 1 \leq x$. Similarly

$$k + 3x + p + t = \max(k + 3x, 2k - \gamma + 2x - y, 2k - \gamma + 2x - z, 3k - 2\gamma + 2x - S) \leq 3T :$$

the first term is at most $\frac{5}{2}k$; the middle two are at most $2k - \gamma + 2 \lfloor k/2 \rfloor - J = 3T$; and the last is at most $3k - 2\gamma + 2 \lfloor k/2 \rfloor - (2T - k) = 6k - 4T \leq 3T$, using $\gamma = T - \lfloor k/2 \rfloor$. Finally $y + z + 2\lambda \leq 4\lambda \leq T$.

Case 3: $S > T$. Apply Lemma 4.5 with $\text{Gap}(\lambda, \gamma)$ and $\Delta = S - T$. Write $\epsilon = k - 2 \lfloor k/2 \rfloor \in \{0, 1\}$ and note $\gamma - h = 2T - k - \lfloor k/2 \rfloor$. Using $y, z \leq \lambda \leq \gamma - h$ and $x \leq \lfloor k/2 \rfloor$,

$$\begin{aligned} x + y &\leq \lfloor k/2 \rfloor + \gamma \leq T, & k - x - z + \Delta &= h + y \leq \gamma, \\ k - x - y &= h + z - \Delta < h + \lambda \leq \gamma, \\ 2x + 2\Delta + k - y - z &= 4x + y + z + k - 2T \leq 4 \lfloor k/2 \rfloor + 2(\gamma - h) + k - 2T = 2T - 3\epsilon, \\ y + z + 2\lambda &\leq 4\lambda \leq T, \\ 2k + x - y + \lambda + \Delta &= 2k + 2x + z + \lambda - T \leq 2k + 2 \lfloor k/2 \rfloor + 2(\gamma - h) - T = 3T - 2\epsilon. \end{aligned}$$

□

7 Proof of the main theorem

Proof of Theorem 1.1. For $2 \leq k \leq 6$ see the paragraph after Theorem 1.1. Let $k \geq 7$ and suppose $\tau(\mathcal{H}) > T = \lceil 6k/7 \rceil$. If $\beta \geq \lfloor k/2 \rfloor$, Proposition 6.2 shows that no pair intersection lies in $(\alpha, k/2]$. Otherwise Propositions 6.2 and 6.4 show the same. Since $\alpha = \lfloor T/2 \rfloor$, every pair intersection is at most $T/2$ or greater than $k/2$, and Theorem 5.2 gives $\tau(\mathcal{H}) \leq T$, a contradiction.

Finally, let $k \geq 2$. For families of nonempty sets of size at most k , add to each edge new points private to that edge until it has k points. A transversal of the original family is one of the padded family. Conversely, replacing each private point of a transversal of the padded family by an arbitrary point of the corresponding original edge gives a transversal of the original family of no larger size. So padding preserves the transversal number, and it preserves property (7, 2) since a two-point transversal of original edges meets the padded edges. □

8 Where six-sevenths comes from

We record the three inequalities that determine the coefficient and indicate why they do not appear to be relaxable by changing parameters. Write $t = T/k$ and normalize sizes by k .

Static allocation. Corollary 3.2 with $a = T/2$ needs $3 + t/2 \leq 4t$, that is $t \geq 6/7$; this is the load of the Hall allocation, in which the distinguished cell lies in three of the four requests and every private point in one.

Starting the first stage. A request of T points built from two edges that meet in q points leaves $2 - t - q$ of their private points available to the response. Both traces can be forced to be small (at most $t - 1/2$) only if $2 - t - q \leq 2t - 1$, that is $q \geq 3 - 3t$. The first stage must start where the finishing argument stops, at $t/2$, so $3 - 3t \leq t/2$, again $t \geq 6/7$.

The two-medium triples. In Proposition 5.1, a triple with two medium cells and a third cell $z \leq 1 - t$ is closed by the near-core construction, whose budget includes $2 + M + z \leq 3t$ (in the alternative $S > T$ of Lemma 4.7); with $M = t/2$ and $z = 1 - t$ this is once more $t \geq 6/7$. Cells with $z > 1 - t$ are closed by the two-cores construction, whose first request needs $z \geq 1 - t$.

At $t = 6/7$ the three thresholds coincide: $4t - 3 = t/2 = 3 - 3t = 3/7$. For $t < 6/7$ they separate, and the window $4t - 3 < M < 3 - 3t$ contains two kinds of problematic configurations.

- If $M \leq t/2$, the balanced request of Lemma 2.4 applied to a pair attaining the largest pair intersection M of size at most $k/2$ (the M of Theorem 5.2) can return triples with cells $M \geq y \geq z$, $t - \frac{1}{2} < y \leq 1 - \frac{t+M}{2}$ and $3t - 2 - M < z < 1 - t$; for $5/6 < t < 6/7$ these conditions are compatible for every M in the window with $M \leq t/2$. For them both alternatives of Lemma 4.7 fail, as does Lemma 4.6, in the orientations used in Proposition 5.1.
- If $t/2 < M < 3 - 3t$, the first stage is not available, and any request of T points built from two edges meeting in M points leaves $2 - t - M > 2(t - \frac{1}{2})$ of their private points, so the response can create a medium cell next to the cell of size $M > t/2$.

Exploratory computations, not used in any proof, are consistent with this picture. They used continuous relaxations of the covering problems (cell sizes normalized by k , request contents chosen by mixed integer programming, adversarial responses bounded by column generation), at $t = 0.855$. Allowing oneself to discard one edge after the fourth response and to close the resulting new good triple statically (the device used in the small-maximum case of Theorem 5.2) closed the tested configurations of the first kind. For configurations of the second kind, the one-response constructions we tested, with or without this device, required a budget above 0.855 at every tested point. Thus the obstruction moves but, as far as these experiments show, reappears at the same coefficient. An improvement of $6/7$ seems to require a genuinely new ingredient for triples that combine a cell slightly above $t/2$ with a medium cell. The conjectured value $3/4$ remains open.

A Proofs of the adaptive closures

We use the notation of Section 2. In each proof, K is the response to the first request, and the three final requests are described by their *bases*, sets that they must contain; remaining points are then distributed by Lemma 2.2. By Lemma 2.3 it suffices to cover the candidate pairs listed there. When no triple cell survives, these are the three *complementary products*

$$(X \setminus K) \times (K \cap P_G), \quad (Y \setminus K) \times (K \cap P_F), \quad (Z \setminus K) \times (K \cap P_E).$$

Proof of Lemma 4.1. Request K avoiding X , Y and a set $Z_0 \subseteq Z$ of $\min(z, T - x - y)$ points; this is legal as $T \geq x + y$. Put $Q = Z \cap K$, $A = (F \cap K) \setminus Q$, $B = (G \cap K) \setminus Q$, $C = E \cap K$, $V = Z \setminus Q$

and $D = E \setminus (X \cup Y \cup C)$, with sizes q, a, b, c, v, d . Then $A \subseteq P_F$, $B \subseteq P_G$, $C \subseteq P_E$, $q \leq (S - T)_+$, $q + a + b + c \leq k$,

$$a \leq k - x - z, \quad b \leq k - y - z, \quad c \leq k - x - y, \quad d = k - x - y - c.$$

By Lemma 2.3 the candidate pairs are $Q \times E$, $X \times B$, $Y \times A$ and $V \times C$. We make all three final requests contain Q and their union contain E , and cover the other three products by bases.

Since $G \cap K = Q \cup B$ avoids $Y \cup Z_0$, we have $q + b \leq k - y - |Z_0|$, so $q + x + b \leq \max(k + x - y - z, k + 2x - T) \leq T$; similarly $q + y + a \leq \max(k - x + y - z, k + 2y - T) \leq T$. Also $z \leq 4T/5 \leq T$, as $4T \geq 2k + 3z \geq 5z$.

Case 0: $c \leq T - z$. Use bases $Q \cup X \cup B$, $Q \cup Y \cup A$, $Z \cup C$; each has at most T points. Their total size plus d is $k + 2q + a + b + z$. Using $a + b \leq 2k - x - y - 2z$ and $q \leq (S - T)_+$, this is at most $\max(3k - S, 3k + S - 2T) \leq 3T$. Distribute D among the residual capacities (Lemma 2.2). All candidate pairs are covered: $Q \subseteq Z$ lies in all three requests, and $E = X \cup Y \cup C \cup D$ lies in their union.

In the remaining cases $c > T - z$. Put $a_0 = k + x + z - 2T$ and $b_0 = k + y + z - 2T$.

Case 1: $a < a_0$. Split $C = C_2 \sqcup C_3$ with $|C_2| \leq T - y - a - z$ and $|C_3| \leq T - z$, and use bases $Q \cup X \cup B$, $Y \cup A \cup Z \cup C_2$, $Z \cup C_3$. The split is possible because $y + a + z < k + x + y + 2z - 2T \leq T$ and $a + c < (k + x + z - 2T) + (k - x - y)$, so $c < (T - y - a - z) + (T - z)$ by $2k + 3z \leq 4T$. The total size plus d is $k + q + a + b + 2z \leq 2k + 2z - c < 2k + 3z - T \leq 3T$; distribute D . The products $X \times B$ and $Y \times A$ are covered by the first two bases, $V \times C$ by the last two (both contain $Z \supseteq V$), and $Q \times E$ as before.

Case 2: $b < b_0$. Interchange the roles of (X, B, x, b) and (Y, A, y, a) in Case 1.

Case 3: $a \geq a_0$ and $b \geq b_0$. Put $u = c + z - T$, so $0 < u \leq c$. Choose $C_{12} \subseteq C$ with $|C_{12}| = u$, put $C_3 = C \setminus C_{12}$, and choose an integer θ with

$$\max(0, y + a + z + u - T) \leq \theta \leq \min(v, T - q - x - b - u).$$

Such θ exists: $q + b + c \leq k - a \leq 2T - x - z$ gives $T - q - x - b - u \geq 0$; $q + a + c \leq k - b \leq 2T - y - z$ gives $y + a + z + u - T \leq v$; and $y + a + z + u - T \leq T - q - x - b - u$ is equivalent to $q + a + b + 2c + x + y + 3z \leq 4T$, which follows from $q + a + b + c \leq k$, $c \leq k - x - y$ and $2k + 3z \leq 4T$. Split $V = V_{13} \sqcup V_{23}$ with $|V_{13}| = \theta$ and use bases

$$Q \cup X \cup B \cup C_{12} \cup V_{13}, \quad Q \cup Y \cup A \cup C_{12} \cup V_{23}, \quad Z \cup C_3.$$

The choice of θ bounds the first two by T (the second has $q + y + a + u + v - \theta = y + a + z + u - \theta$ points, as $q + v = z$), and the third has exactly T points. The total size plus d is $k + q + a + b + 2z + u \leq 2k + 3z - T \leq 3T$; distribute D . The first two bases cover $X \times B$, $Y \times A$ and $V \times C_{12}$, the third covers $V \times C_3$, and $Q \times E$ is covered as before. $\square$

Proof of Lemma 4.2. Request K avoiding $X \cup Y \cup Z$ and $T - S$ points of P_F ; this is possible as $0 \leq T - S \leq k - x - z$. No triple cell survives. Put $A = F \cap K$, $B = G \cap K$, $C = E \cap K$, of sizes a, b, c ; then $a \leq k - T + y$, $b \leq k - y - z$ and $a + b + c \leq k$. The candidate pairs are $X \times B$, $Y \times A$, $Z \times C$.

If $a \leq T - y - z$, split $B = B_1 \sqcup B_2$ with $|B_i| \leq T - x - z$ (possible since $2(T - x - z) \geq k - y - z \geq b$) and use bases $X \cup Z \cup B_1$, $X \cup Z \cup B_2$, $Y \cup Z \cup A$. Their total size plus c is at most $k + 2x + y + 3z \leq 3T$, so C can be distributed. As every request contains Z , all three products are covered.

If $a > T - y - z$, then $b + c \leq k - a < k - T + y + z \leq 2(T - x - z)$. Split $B \cup C$ into two parts of size at most $T - x - z$ and use $X \cup Z$ together with one part, twice, and $Y \cup A$, of size at most $k - T + 2y \leq T$. $\square$

Proof of Lemma 4.3. Request K avoiding $X \cup Y \cup Z$ and $p = \max(0, k - 2(T - x) - y - z)$ points of P_G . This fits ($p \leq k - y - z$ as $T \geq x$), and the request has $\max(S, k + 3x - 2T) \leq T$ points. No triple cell survives; with $A = F \cap K$, $B = G \cap K$, $C = E \cap K$ we have $b \leq 2(T - x)$, $c \leq k - x - y$ and $a + b + c \leq k$.

If $b > k + y + z - T$, split B into two parts of size at most $T - x$ and use X with each part, and $Y \cup Z \cup A \cup C$, of size at most $y + z + k - b < T$.

Otherwise $b \leq k + y + z - T \leq 2(T - x - y)$, the last step being $3T \geq k + 2x + 3y + z$. Split $B = B_1 \sqcup B_2$ with $|B_i| \leq T - x - y$ and use bases $X \cup Y \cup B_1$, $X \cup Y \cup B_2$, $Y \cup Z \cup C$; the last has at most $k - x + z \leq T$ points. Their total plus a is at most $k + 2x + 3y + z \leq 3T$; distribute A . Every request contains Y , which covers $Y \times A$. $\square$

Proof of Lemma 4.4. Request $X \cup Y \cup Z$, p points of P_E , t points of P_F , and $T - S - p - t$ points of P_G ; these fit, as $u \geq 0$ and $T \leq k$. No triple cell survives. The trace of K on E has at most $k - x - y - p \leq u$ points, hence at most ℓ ; likewise on F . Its trace B on G has at most $k - T + x + p + t$ points. Split B into two parts of sizes at most $\lceil (k - T + x + p + t)/2 \rceil$ and use bases $X \cup B_1$, $X \cup B_2$ and $Y \cup Z \cup (F \cap K) \cup (E \cap K)$. The first two have at most T points because $k + 3x + p + t \leq 3T$ (all quantities are integers), and the third has at most $y + z + 2\ell \leq T$. $\square$

Proof of Lemma 4.5. Request K avoiding X, Y and $\min(z, T - x - y)$ points of Z . Put $P = Z \cap K$, $A = (F \cap K) \setminus P$, $B = (G \cap K) \setminus P$ and $C = E \cap K$, with sizes p, a, b, c ; then $p \leq \Delta$ and $b \leq k - y - z$. Moreover $|E \cap K| = c \leq k - x - y \leq u$ and $|F \cap K| = a + p \leq k - x - z + \Delta \leq u$, so $c \leq \ell$ and $a + p \leq \ell$ by the gap. The candidate pairs are $P \times E$, $X \times B$, $Y \times A$ and $(Z \setminus P) \times C$.

Split $B = B_1 \sqcup B_2$ nearly equally and use bases $X \cup P \cup B_1$, $X \cup P \cup B_2$ and $Y \cup Z \cup A \cup C$. The first two have at most $x + \Delta + \lceil (k - y - z)/2 \rceil \leq T$ points, and the third at most $y + z + 2\ell - p \leq T$. All three contain P . Distribute $D = E \setminus (X \cup Y \cup C)$, so that the union contains E ; the total load is

$$k + x + z + 2p + a + b \leq k + x + z + p + \ell + b \leq 2k + x - y + \ell + \Delta \leq 3T.$$

$\square$

Proof of Lemma 4.6. Request Y , g points of X , g points of Z , and further points of F so that the request contains exactly $T - y$ points of F , taking remaining points of $X \cup Z$ first. This is possible as $2g \leq T - y \leq k$. Put $P = X \cap K$ and $R = Z \cap K$, of sizes p and r ; by the priority rule $p + r \leq (S - T)_+$. As K avoids Y , these are the only triple cells. Let $A = (F \cap K) \setminus (P \cup R)$, $B = (G \cap K) \setminus R$, $C = (E \cap K) \setminus P$, with sizes a, b, c ; then $d := |F \cap K| = p + r + a \leq k - T + y$. The traces of K on E and on G have at most $k - y - g \leq u$ points, hence at most ℓ .

Use bases $X \cup (G \cap K)$, $Z \cup (E \cap K)$ and $Y \cup (F \cap K)$, of sizes at most $x + \ell$, $z + \ell$ and $k - T + 2y$, all at most T . Each contains P and R . Distribute $D = P_E \setminus K$ and $W = P_G \setminus K$, of sizes $k - x - y - c$ and $k - y - z - b$, so that the union contains $E \cup G$. The total load is

$$2k - y + 2(p + r) + a = 2k - y + (p + r) + d \leq 3k - T + (S - T)_+ \leq 3T,$$

by $3k \leq 4T$ if $S \leq T$ and by $3k + S \leq 5T$ otherwise. The candidate pairs are $P \times G$ and $R \times E$, covered since P, R lie in every request and the union contains $E \cup G$, and $(X \setminus P) \times B$, $(Z \setminus R) \times C$, $Y \times A$, covered by the three bases. $\square$

Proof of Lemma 4.7. Request K avoiding $Y \cup Z$ and $\min(x, T - y - z)$ points of X . The only possible triple cell is $P = X \cap K$, of size $p \leq \Delta$. Put $X' = X \setminus P$, $A = K \cap P_E$, $B = K \cap P_F$, $C = K \cap G$ and $W = P_G \setminus K$, with sizes a, b, c, w ; then $C \subseteq P_G$,

$$a \leq k - x - y, \quad b \leq k - x - z, \quad c + w = k - y - z.$$

By Lemma 2.3 the candidate pairs are $P \times (Y \cup Z \cup C \cup W)$, $X' \times C$, $Y \times B$ and $Z \times A$.

Case 1: $p + a \leq m$. Put P in all three requests, X' in requests 1 and 3, $Y \cup B$ in request 1, $Z \cup A$ in request 2; distribute C between requests 1, 3 and W among all three. The loads before distribution are $x + y + b \leq k + y - z \leq T$, $p + a + z \leq m + z \leq T$ and $x \leq T$. The distribution of C is possible since $2x + y + b + c \leq 2k + x - 2z \leq 2T$, and then that of W since the total load is $2x + k + p + a + b \leq 2k + x + m - z \leq 3T$; as W may go to any request, these two inequalities are all the conditions of Lemma 2.2. Every candidate pair is covered.

Case 2: $p + a > m$. Then $|E \cap K| = p + a > k/2$ by the dichotomy (this includes the possibility $K = E$), so $|G \cap K| = c < k/2$; in particular $K \neq G$, and the dichotomy gives $c \leq m$. Put P in requests 1, 2; X' and C in request 2; $Y \cup B$ in request 1; Z in requests 1, 3; and distribute A between requests 1, 3 and W between requests 1, 2. The initial loads are $p + y + z + b \leq \Delta + k - x + y \leq T$, $x + c \leq x + m \leq T$ and $z \leq T$. The three conditions of Lemma 2.2 for A and W follow from

$$\begin{aligned} (p + y + z + b) + z + a &\leq \Delta + 2k - 2x + z \leq 2T, \\ (p + y + z + b) + (x + c) + w &= p + x + k + b \leq \Delta + 2k - z \leq 2T, \\ p + y + 2z + b + x + c + a + w &= p + x + k + z + a + b \leq \Delta + 3k - x - y \leq 3T. \end{aligned}$$

Every candidate pair is covered: P meets Y and Z in request 1, C in request 2 and W in request 1 or 2; X' and C share request 2; Y and B share request 1; and both allowed labels of A meet the label $\{1, 3\}$ of Z .

If $S \leq T$, the first request contains X , so $P = \emptyset$ and $\Delta = 0$, and W has no candidate partner. Case 1 then needs only the first distribution, and Case 2 only the distribution of A , whose conditions are $k - x + y \leq T$ and $2k - 2x + z \leq 2T$. $\square$

Proof of Lemma 4.8. Put $Y = E \cap G$, $Z = F \cap G$, $Y' = F \cap H$, $Z' = E \cap H$ and $U = Y \cup Z \cup Y' \cup Z'$; the six pair cells are disjoint. Write $\sigma = |Y| + |Z|$ and $\rho = |Y'| + |Z'|$, both at most $2m$, and put

$$p = \lceil k/2 \rceil - \min(\sigma, \rho), \quad q = T - \lceil k/2 \rceil - \max(\sigma, \rho).$$

Both are nonnegative; $p \leq \lceil k/2 \rceil \leq b$ and $q \leq \lfloor k/2 \rfloor < x$. Choose $B_0 \subseteq B$ and $X_0 \subseteq X$ of sizes p, q , and request an edge I avoiding $U \cup B_0 \cup X_0$, a set of exactly T points. The trace of I on G has at most $k - \sigma - p = \lfloor k/2 \rfloor + \min(\sigma, \rho) - \sigma \leq \lfloor k/2 \rfloor$ points, and similarly on H ; by the dichotomy both are at most m . In particular $|B \cap I| \leq m$, and $|X \cap I| \leq x - q$.

Choose $B_1 \subseteq B$ with $B \cap I \subseteq B_1$ and $|B_1| = \min(b, T - x)$, possible since $T - x \geq m$. Use the requests $X \cup B_1$ and $(X \cap I) \cup (B \setminus B_1)$. The first has at most T points; the second has $|X \cap I| + \max(0, b - T + x)$ points, and $x - q + b - T + x \leq T$ follows from $q \geq T - \lceil k/2 \rceil - 2m$ and the last hypothesis.

As all triple intersections are empty, the candidate pairs of E, F, G, H lie in $X \times B$, $Y \times Y'$ or $Z \times Z'$. The last two products lie in U , which I avoids. A pair in $X \times B_1$ misses the response to $X \cup B_1$. A pair $\{u, v\}$ with $u \in X$ and $v \in B \setminus B_1$ has $v \notin I$, so it meets I only if $u \in I$; then it misses the response to $(X \cap I) \cup (B \setminus B_1)$. The seven edges E, F, G, H, I and the two responses form a bad subfamily. $\square$

Research assistance

This work made substantial use of AI systems. Claude (Anthropic) and Codex (OpenAI) were used for mathematical exploration, for developing and checking the local constructions and the case analysis, for computational experiments that guided the choice of constructions, for literature

searches, and for preparing the manuscript. The proofs in this paper are hand arguments with exact integer quantities; no computation is needed to verify them. [The human authors must describe here the independent verification they performed.]

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